ERGA Assembly Report

v24.10.15

Tags: ERGA-BGE

TxID	3078427
ToLID	**icHydHamu2**
Species	Hydroglyphus hamulatus
Class	Insecta
Order	Coleoptera

Genome Traits	Expected	Observed
Haploid size (bp)	398,697,633	475,330,853
Haploid Number	17 (source: ancestor)	15
Ploidy	2 (source: ancestor)	2
Sample Sex	X0	X0

EBP metrics summary and curation notes

Obtained EBP quality metric for collapsed: 6.7.Q22

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

. Observed Haploid Number is different from Expected

. QV value is less than 40 for collapsed
. Kmer completeness value is less than 90 for collapsed
. Assembly length loss > 3% for collapsed

Curator notes

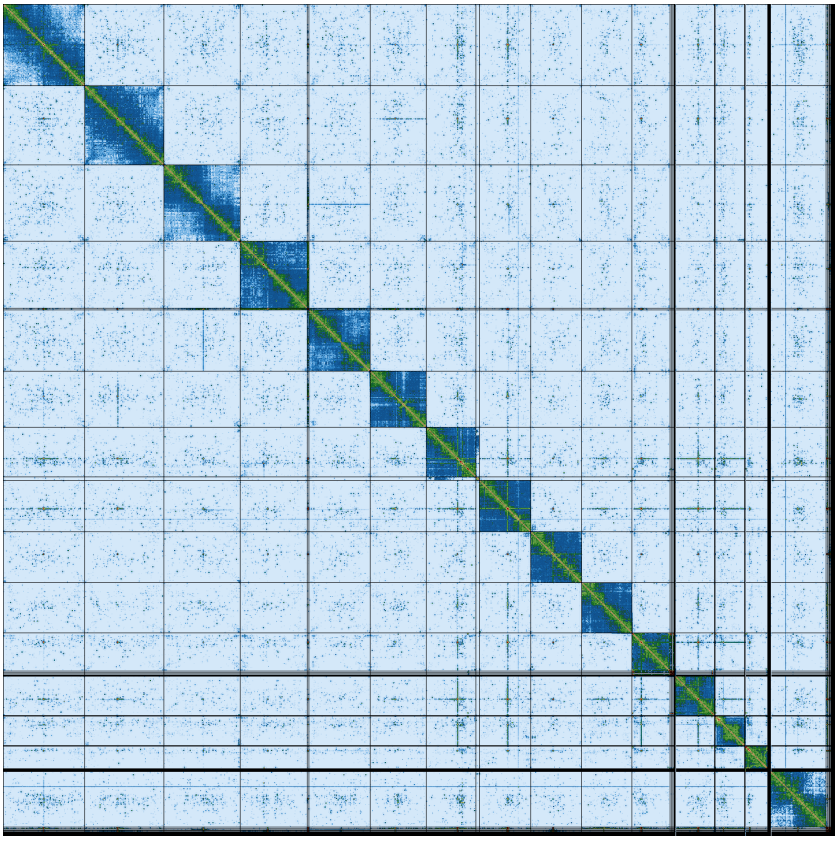
. Interventions/Gb: 223
. Contamination notes: "Some contaminant contigs were detected by FCS-GX and were removed (Wolbachia and other proteobacteria). 3 Ciliophora contigs were detected in the blobtools plot and were removed, as well as most of the no-hit contigs with similar coverage/GC in the blob plot and similar profile in the pretext map."
. Other observations: "The assembly was produced with hifiasm using HiFi data. The resulting collapsetd *.bp.p_ctg.gfa was selected, screened for contamination with FCS-GX, and purged of haplotigs with purge_dups. Scaffolding was performed with YaHS. 80 changes were made using PretextView. The QV value comes from merkuryfk and is low (30). However the QV value estimated by merkury is 62. I don't know why there is such a big difference."

Quality metrics table

Metrics	Pre-curation collapsed	Curated collapsed
Total bp	504,646,154	475,330,853
GC %	34.69	33.86
Gaps/Gbp	576.64	641.66
Total gap bp	58,200	61,000
Scaffolds	195	91
Scaffold N50	28,882,811	32,287,851
Scaffold L50	7	6
Scaffold L90	17	13
Contigs	486	396
Contig N50	2,434,142	2,570,636
Contig L50	61	57
Contig L90	235	200
QV		22.7
Kmer compl.		82.47
BUSCO sing.	96.6%	97.0%
BUSCO dupl.	1.6%	1.1%
BUSCO frag.	0.2%	0.2%
BUSCO miss.	1.6%	1.7%

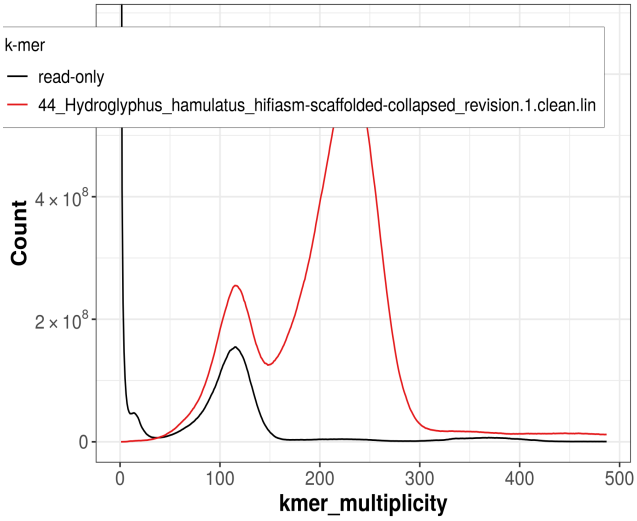
BUSCO: 6.0.0 (euk_genome_min, miniprot) / Lineage: insecta_odb12 (genomes:79, BUSCOs:3114)

HiC contact map of curated assembly

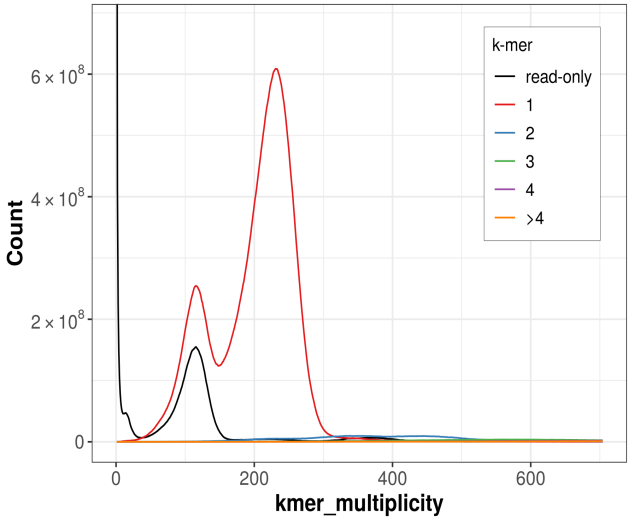


collapsed [LINK]

K-mer spectra of curated assembly

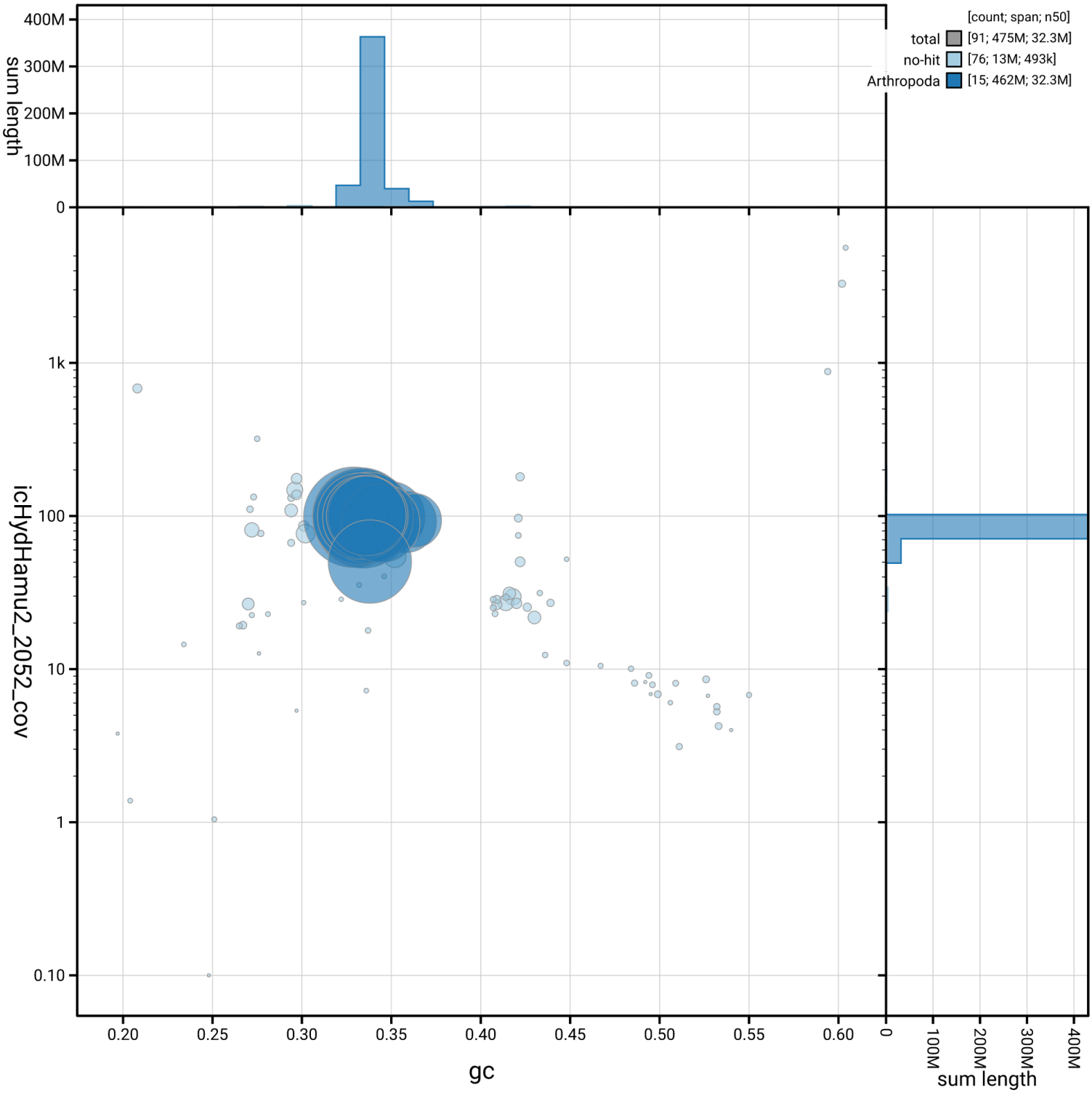


Distribution of k-mer counts coloured by their presence in reads/assemblies



Distribution of k-mer counts per copy numbers found in asm

Post-curation contamination screening



collapsed. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

Data profile

Data	PacBio HiFi	Arima HiC
Coverage	NA	NA

Assembly pipeline

- **hifiasm**
    |_ *ver:* 0.25.0
    |_ *key param:* NA
- **purge_dups**
    |_ *ver:* 1.2.6
    |_ *key param:* NA
- **YaHS**
    |_ *ver:* 1.2a.2
    |_ *key param:* NA
- **NBIS/Earth-Biogenome-Project-pilot**
    |_ *ver:* 85e04ded1d
    |_ *key param:* NA
- **sanger-tol/blobtoolkit**
    |_ *ver:* 0.10.0
    |_ *key param:* NA

Curation pipeline

- **sanger-tol/curationpretext**
    |_ *ver:* 1.5.1
    |_ *key param:* NA

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